



Team CodeHackers

Round: II

IDEA SUBMISSION

Echo
Sphere

Team Details:

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Track Chosen: Adaptive AI Sales and Negotiation Agent

Problem Statement

Build a Real-time voice AI sales agent that conducts complete customer qualification and sales conversations, adapts to objections and changing requirements, answers product and pricing questions using verified information, integrates with CRM or calendar systems, and escalates to a human with full conversation context.



Problem Description:

Commercial auto drivers in India must choose an electric three-wheeler that determines income, yet vehicle specifications, EMI options, charging access, service coverage, and incentives are fragmented across multiple sources. Static sales scripts cannot handle changing requirements, regional languages, interruptions, or trust-sensitive questions, causing confusion, delayed decisions, and lost EV adoption.

Why it matters:

An electric three-wheeler is not only a vehicle; it is a driver's earning asset. A poor purchase decision can increase financial pressure, reduce daily uptime, and weaken confidence in India's electric-mobility transition. Financing and adoption barriers are documented by RMI and Government of India sources

Echo Sphere





Proposed Solution / Idea Overview:

SaarthiEV is a multilingual, real-time voice sales and negotiation agent for India's commercial electric three-wheeler market. It speaks with a driver in a preferred Indian language, discovers route, daily distance, passenger load, fuel spend, budget, down payment, and charging access, and maintains this context as the conversation changes.

Instead of following a fixed script, the agent handles interruptions, remembers earlier objections, retrieves approved vehicle, EMI, charging, service, and incentive information, compares options under visible assumptions, and proposes a verified next action: test drive, dealer meeting, lender callback, or human escalation. Agora provides the live RTC channel, natural turn-taking, interruption handling, transcripts, session context, and observable handoff.

Our unique contribution is a structured livelihood-and-vehicle memory that converts an unstructured voice conversation into a trustworthy, action-ready lead brief for drivers and EV partners.





Additional Notes

Prototype scope: The hackathon MVP focuses on one realistic driver journey: the user asks about EMI, changes the daily route and passenger-load requirement, compares two approved electric three-wheelers, asks about incentives, and ends with a test-drive booking or human callback.

Planned demonstration: The agent will support English plus one Indian language, handle natural interruptions, update a visible requirement state, call controlled vehicle/EMI/route tools, show source-backed answers, and create a structured CRM-style handoff brief.

Responsible AI boundary: SaarthiEV will provide indicative calculations and verified information only. It will not approve loans, guarantee subsidies, certify vehicle suitability, promise income or savings, or replace authorized lenders, dealers, service teams, or engineers.

Success measures: We will track response latency, qualification completeness, interruption recovery, tool-call reliability, next-action completion, human handoff quality, and unsupported-answer rate. Prototype numbers will be labeled as observed results or pilot targets.

Scale path: The product can expand from one dealer to OEMs, NBFCs, charging operators, fleet platforms, additional Indian languages, and multi-city deployments.



TOOLS + MEMORY

Tool calling makes the advice specific—and keeps it honest

Every meaningful recommendation comes from an approved tool or verified knowledge source.



Vehicle catalog

Approved models, price, payload, range, warranty.



EMI scenario

Down payment, tenure, indicative repayment.



Route economics

Energy/fuel cost under stated assumptions.



Charging + service

Verified points near the driver's route.



Policy verification

Current state/category incentive information.



Booking + CRM

Test drive, callback, and timestamped brief.



LIVE MEMORY OBJECT

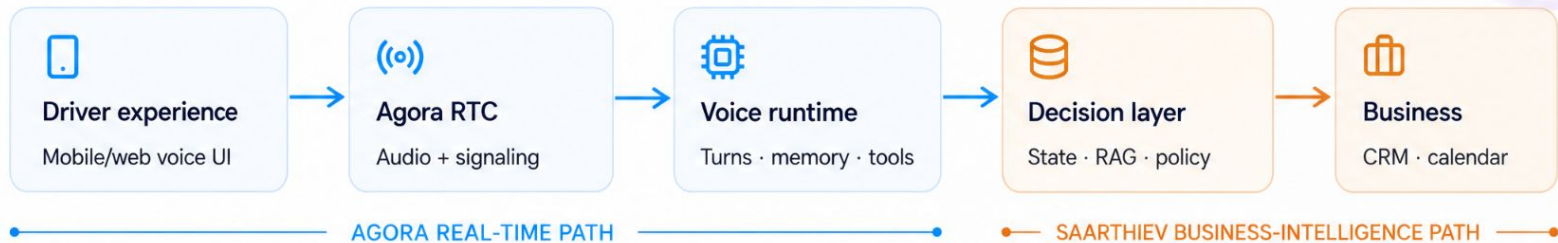
city • route_km • load • EMI_limit • down_payment • shortlist • charging_access • policy_question • objections • confidence • next_action

No unsupported claims leave the system: facts are retrieved, estimates show assumptions, uncertainty triggers escalation.

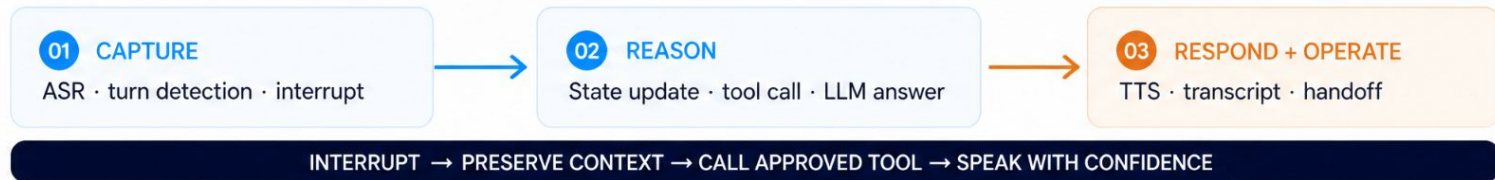
TECHNICAL ARCHITECTURE

Agora carries the live conversation from audio to action

Agora owns the real-time session; SaarthiEV owns the domain decision logic and business workflow.



LIVE CALL FLOW / GROUPED



Business server starts/stops the agent; transcript callbacks, session history, and turn analytics make outcomes observable.[4] [5] [6] [7]

PLANNED AGORA UTILIZATION

Every Agora capability maps to visible product behavior.

Agora is not the microphone layer. It is the real-time product advantage.



Agora technology	SaarthiEV behavior	Judge-visible proof
RTC SDK	Driver and agent share a live voice channel	Natural real-time conversation.
Conversational AI Engine	Agent listens, reasons, speaks, and manages session	End-to-end voice agent.
Interruption handling	Driver interrupts without waiting	Agent stops and recovers naturally.
Custom LLM / RAG	Retrieves vehicle, finance, charging, policy facts	Source-backed answer.
Signaling	Transcript and state update live	Shared driver/operator context.
Live transcripts	Driver and human see the conversation	Transparent handoff.
History + webhooks	CRM receives outcome and context	Operational workflow created.
Runtime metrics	Measures latency, turns, errors, and tools	Production readiness.



Interruptions · live context · observable handoff · measurable latency.

Agora supports live transcript callbacks, interrupted status, session history, and turn-level analytics.

A focused MVP can prove the complete adoption journey



HACKATHON SCOPE

One commercial auto driver is worried about EMI and range, compares two models, asks about subsidy, and books a test drive or human callback.



PROTOTYPE COMPONENTS

- Agora RTC voice • one Indian language
- controlled vehicle catalog • mock EMI tool
- route economics • charging/service directory
- CRM brief • calendar action



BUILD SEQUENCE



ROADMAP

Controlled demo → partner APIs → live integrations → multi-city rollout



Acceptance test: 3 requirement changes → 1 verified next action



EXPLICIT BOUNDARY

No final loan approval, subsidy approval, engineering certification, guaranteed savings, or guaranteed income.

10 / IMPACT + WHY THIS CAN WIN

SaarthiEV makes India's EV transition actionable—one conversation at a time

A responsible pilot measures conversion and safety together—not one at the expense of the other.


< 60 sec

first useful response


80%+

qualification complete


3+

requirement changes handled


0

fabricated commitments



Proposed prototype/pilot targets—not claimed results.



TRUST BOUNDARY



Estimates show assumptions; they are not guarantees.



Policy uncertainty triggers verification or escalation.



Lending and technical approval stay with experts.



Human handoff includes transcript, state, objections, and confidence.

INNOVATION • RELEVANCE • FEASIBILITY • IMPACT • AGORA-NATIVE



A safer, more informed livelihood decision—in the language and moment that works for the driver.

SaarthiEV

Adaptive multilingual voice sales agent for India's electric auto transition

AGORA LIVE

Hindi + English



SaarthiEV Voice Session

AGORA LIVE

Listening — interrupt anytime



Hindi + English



03:21

Live conversation

D Driver: Bhai, EMI 8,000 se zyada nahi honi chahiye. 10:42 AM

A Agent: Samajh gaya. Aapka route roz 90 km hai, right? 10:42 AM

D Driver: Haan, aur charging depot par hi milegi. 10:43 AM

A Agent: Then I'll compare models with depot charging and keep the EMI cap. 10:43 AM

Requirement updated

Vehicle catalog

EMI scenario

Charging lookup

Live requirement state

Route
90 km/day

EMI cap
₹8,000

Charging
Depot

Shortlist
2 models

Next
Test drive


Book test drive / Human callback

Indicative guidance — final approval stays with experts.